A Tribranch Cascaded Context-Aware Fusion Network for Multispectral Object Detection in Remote Sensing

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***Abstract*—Object detection in multispectral remote sensing imagery is essential for practical applications such as environmental monitoring, urban security, emergency response, and precision agriculture. By combining RGB and infrared (IR) information, multispectral sensing effectively mitigates challenges caused by adverse illumination, haze, and complex weather conditions. However, directly fusing heterogeneous spectral modalities remains difficult due to their fundamentally different noise patterns, spatial resolutions, and semantic characteristics. To address these inconsistencies, we propose a tribranch cascaded context-aware fusion network (TCFNet) for multispectral object detection in remote sensing. TCFNet is built upon three complementary branches: two modality-specific branches designed to independently model RGB and IR feature distributions, and a hybrid branch that explicitly captures RGB–IR interactions from the early stages. To effectively integrate multispectral cues, we introduce a cascaded context-aware multiscale fusion (CCMF) module, which progressively aggregates semantic, spatial, and contextual information across different scales. This cascaded design enhances the network's ability to suppress background noise, preserve small-object features, and leverage cross-modal complementarity. Finally, a feature-level multispectral feature fusion (MFF) module synthesizes predictions from all branches to produce robust detection results. Extensive experiments are conducted on four remote sensing datasets, primarily the DroneVehicle and VEDAI. The proposed method demonstrates effectiveness in remote sensing object detection, achieving mAP@50 improvements ranging from 2.4% to 3.5% compared to existing state-of-the-art methods.**

Received 19 November 2025; revised 6 April 2026 and 4 June 2026; accepted 23 June 2026. Date of publication 26 June 2026; date of current version 27 July 2026. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62402022 and in part by the Joint Funds of the National Natural Science Foundation of China under Grant U23A20357. *(Corresponding authors: Jinli Zhang; Yongzhe Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3707553

***Index Terms*—Multispectral fusion, multispectral object detection, remote sensing, RGB–infrared (IR) imagery.**

I. Introduction

MULTISPECTRAL small object detection in remote sensing utilizes information from different spectral bands, such as RGB and infrared (IR), to detect small objects in remote sensing images.

The fact that multispectral remote sensing images incorporate a level of invariance to challenges such as rain and dust means that multispectral small object detection holds significant application value in remote sensing tasks such as pedestrian detection, intelligent traffic supervision, and uncrewed aerial vehicle (UAV) high-altitude detection [2].

Traditional RGB imaging provides rich color and texture information; however, it particularly struggles in the face of challenges such as complex environments, low or oblique illumination, fog, and dust, leading to missed or false detections. Therefore, most RGB-based methods further enhance detection performance by focusing on local discriminative features through attention mechanisms [3]. IR imaging is robust to some of these obstacles and can capture images at night under low-light conditions by leveraging the reflection of moonlight and atmospheric glow, enabling all-weather target detection [4]. In addition, the longer wavelength of infrared light helps mitigate the effects of fog and other adverse weather conditions on imaging. Detecting small objects with low signal strength remains a challenge within IR images, however, due to their low resolution and contrast. Furthermore, infrared image datasets generally suffer from limited samples [5], [6]. Combined with the low signal-to-noise ratio of infrared data, small object detection in such scenarios faces considerable difficulties [7]. Multispectral object detection aims to leverage the complementary information from different spectral bands [1] to improve final detection accuracy.

The prevailing approach performs multimodal fusion first [8], [9] and then applies an existing state-of-the-art detector.

Most existing multispectral fusion methods can be categorized into data-level [10], [11], feature-level [12], and decision-level approaches [13].

Despite progress, these methods still suffer from missed detections. We observe two main reasons.

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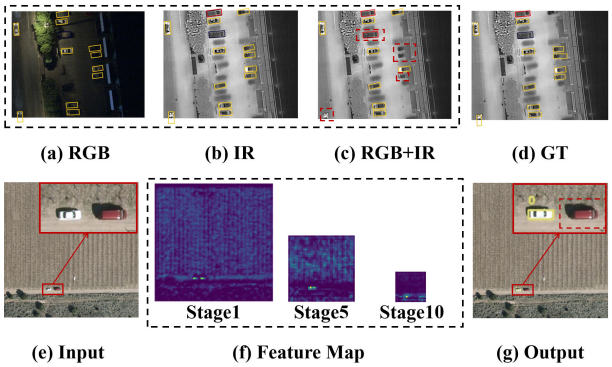


Fig. 1. Multispectral object detection in remote sensing poses two primary barriers: 1) the feature fusion of heterogeneous multispectral data might cause information interference. This is evident in the number of targets missed when performing detection on each channel separately. (a) and (b) Above represent the results of applying YOLOv8 to RGB and IR images, respectively. (c) Presents the results of the multispectral detection model CALNet [1] which fuses features of heterogeneous multispectral, and (d) shows the ground truth and 2) features of small objects usually diminish as the network goes deeper as shown in (e), (f), and (g).

1) The feature fusion of heterogeneous multispectral data might cause information interference. As shown in Fig. 1(a)–(c), CALNet [1], which is meticulously designed for multispectral object detection, misses several targets compared to the union of results obtained via the general object detection model YOLOv8 on each spectral. This poses the request for a more effective fusion strategy.
2) Features of small objects diminish as a network goes deeper, which subsequently leads to missed detections.

To address the above limitations, we propose a tribranch cascaded context-aware fusion network (TCFNet), a simple yet effective architecture designed specifically for multispectral small-object detection in remote sensing.

TCFNet adopts a three-branch design in which an RGB branch and an IR branch independently extract modality-specific representations, while a hybrid RGB–IR branch jointly models cross-spectral interactions. This decoupled-yet-cooperative structure enables each branch to preserve the intrinsic advantages of its corresponding spectral input while simultaneously allowing the hybrid branch to guide cross-modal feature alignment and compensate for spectral inconsistencies. To further enhance robustness, TCFNet introduces a cascaded context-aware multiscale Fusion module that progressively integrates semantic and spatial information across different scales and branches. The cascaded design mitigates heterogeneous fusion interference and preserves discriminative cues of small objects as features propagate to deeper layers. Finally, a feature-level fusion module adaptively aggregates detection predictions from all branches, yielding stable and reliable outputs across diverse environmental conditions.

The separate design of the RGB branch and the IR branch enables the model to easily handle homogeneous data within itself, and the RGB+IR branch learns to predict fusion weights for the outputs of the other branches with awareness of all types of input data. The overall design is extensively verified as effective on three widely used benchmarks with up to 10.12% improvement in terms of the main metric mAP@50 compared to the state-of-the-art.

The main contributions of this article are summarized as follows.

1) We propose a novel TCFNet for multispectral small-object detection in remote sensing scenarios. Unlike existing single-stream or dual-stream fusion approaches, TCFNet explicitly enables modality-specific feature learning and cross-spectral joint reasoning through three independent yet synergistic branches. To the best of our knowledge, this is the first work to explore a tribranch architecture in multispectral remote sensing object detection.
2) We further design a cascaded context-aware multiscale fusion (CCMF) module that performs progressive multiscale integration of intrabranch features across multiscale spaces. This mechanism adaptively enhances semantically weak small-object representations, supports stable hierarchical feature refinement, and introduces only a small number of additional parameters. Benefiting from this design, the proposed network significantly reduces the miss-detection rate of small objects under complex backgrounds or adverse imaging conditions.
3) We introduce an efficient multilevel fusion strategy that encompasses intramodal feature aggregation within single-spectral branches, cross-modal feature fusion within multispectral branches, and a final feature-level multispectral feature fusion (MFF) module, thereby achieving a progressive integration of multispectral features from local to global levels.
4) We conducted experiments on four multispectral remote sensing datasets, including two widely used multispectral remote sensing datasets, DroneVehicle and VEDAI. Results show that TCFNet achieves state-of-the-art performance across multiple key metrics, with mAP@50 improvements of 2.4%–3.5% over the best existing methods. In addition, extensive ablation studies validate the effectiveness of each component and demonstrate the robustness and generalization capability of the proposed model.

II. Related Work

A. Multispectral Object Detection

Traditional object detection primarily relies on RGB images [14]. While these methods capture rich features, they often suffer from poor robustness. Infrared object detection, leveraging the all-weather adaptability and low power consumption of infrared detectors, has been widely used in UAV early warning systems [15] and pedestrian-vehicle detection [16], [17]. However, infrared imaging has limitations, such as lower resolution and a lack of color information, which hinder object recognition and classification. To enhance infrared detection accuracy, researchers have explored multispectral fusion techniques [18], which integrate RGB and IR images to complement each other's weaknesses and improve detection robustness and accuracy.

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Many multispectral fusion methods have demonstrated strong performance, especially in complex environments. For example, Sun et al. [19] introduced the DroneVehicle dataset for multispectral vehicle detection, but its sensor-induced positional discrepancies hinder fusion experiments. To address this, Yuan et al. [11] proposed the TSRA method to align images across different spectra at the data level. In addition, Zhang et al. [18] introduced a halfway feature fusion strategy that iteratively refines multispectral features, significantly improving detection performance.

Recently, numerous advanced feature-level fusion approaches have emerged. Qingyun et al. [8] introduced the cross-modality fusion transformer (CFT), embedding it into the feature extraction backbone to enable intra and intermodality fusion. Shen et al. [20] proposed a dual cross-attention Transformer framework, incorporating an iterative interaction mechanism to mitigate the computational burden of deep Transformer stacking. You et al. [21] developed MSANet, combining Transformer and CNN advantages to enhance feature extraction and context dependency modeling across modalities. Meanwhile, a number of decision-level fusion-based methods have also been proposed. Existing studies overly focus on the exploration of feature-level fusion, while lacking sufficient attention to the excellent performance potential inherent in strategy-level fusion. Liu et al. [22] proposed a prior-information-enhanced mixture-of-experts fusion network (PMFN) for multimodal remote sensing image object detection, which employs a dynamic gating network, frequency-decoupled fusion experts, and a cross-scale channel interleaving strategy to adapt to complex environments. Furthermore, the rapid development of generative models has provided new insights for multispectral image processing. Diffusion models, in particular, have demonstrated superior capabilities in modeling complex data distributions for image restoration and enhancement. Tang et al. [23] proposed a triple-flow RGB-T fusion network to model modality-specific and shared representations, alleviating cross-modal inconsistency. For instance, Yan et al. [24] introduced an efficient image enhancement method by incorporating a diffusion-based frequency before preserving high-frequency details. In more complex scenarios, a unified diffusion-based framework [25] was developed to simultaneously handle deblurring and high-dynamic-range (HDR) imaging in dynamic environments, offering a robust foundation for subsequent object detection tasks. In addition, multiview learning and adaptive fusion have also been applied to multimodal perception [26], proving the efficacy of cross-representation complementary features.

However, most existing methods still rely on a single fusion paradigm, which forces the detector to depend excessively on fused multispectral features and further exacerbates the semantic inconsistency between RGB and IR modalities. To mitigate these issues, we adopt a tribranch design that performs modality-specific learning in two independent branches and explicitly models cross-spectral interactions in a hybrid RGB–IR branch. This architecture enables more flexible, disentangled, and semantically consistent multispectral-representation learning.

B. Multispectral Small Object Detection

With the advancements in UAV and infrared sensor technologies, small object detection in remote sensing has transcended the limitations of the visible spectrum, and researchers are increasingly adopting infrared and multispectral images for this task. Most methods in the previous chapter focused on multispectral detection algorithms that primarily emphasize multispectral feature fusion, lacking dedicated designs for small objects. Therefore, this chapter focuses on introducing methods improved specifically for small objects. Zhang et al. [27] took the resolution of input images as the entry point and proposed SuperYOLO, which improves detection accuracy while reducing computational cost by fusing multimodal data and employing superresolution learning. However, such methods require significant computational resources during training. Zhu et al. [28] introduced multispectral DETR, leveraging deformable attention and spectrum manipulation techniques to enhance multispectral feature extraction, demonstrating strong performance on the VEDAI dataset. Zhang et al. [29] tackled challenges in RGBT person detection with a large-scale dataset, RGBTDronePerson, and proposed QLS and QCE to improve cross-modal feature reliability. Despite these efforts, existing datasets remain insufficient in scale. To address this, Ying et al. [30] developed RGBT-Tiny, a large-scale benchmark for RGBT small-object detection. Liu et al. [66] proposed the DPAL framework, which tackles cross-modal and task-level misalignment in multimodal detection through the CMAM and THAM modules, achieving strong performance on multiple multimodal benchmarks. Their work highlights the importance of alignment learning for enhancing anchor-free detectors in multispectral scenarios. Beyond alignment, the inherent quality of features in degraded or low-contrast environments remains a critical bottleneck for small-object detection. Recent studies have explored specialized architectures to mitigate these effects, such as deep HDR imaging via nonlocal networks [31] to capture long-range dependencies in extreme lighting. In addition, modeling the optical imaging pipeline and learning contrastive-based representations [32] has proven effective for restoring images with hybrid corruptions, thereby providing clearer feature maps for small-object localization.

The above methods have achieved good results on relevant datasets, but they pay excessive attention to the application of multispectral features and lack targeted optimization for the feature enhancement of small objects themselves. Therefore, this article proposes a CCMF module for the multispectral small-object detection task, aiming to enhance the feature representation ability of small objects during the multispectral feature extraction process.

III. Methodology

In this section, we mainly introduce our method, TCFNet. Fig. 2 shows the main architecture of our TCFNet. It consists

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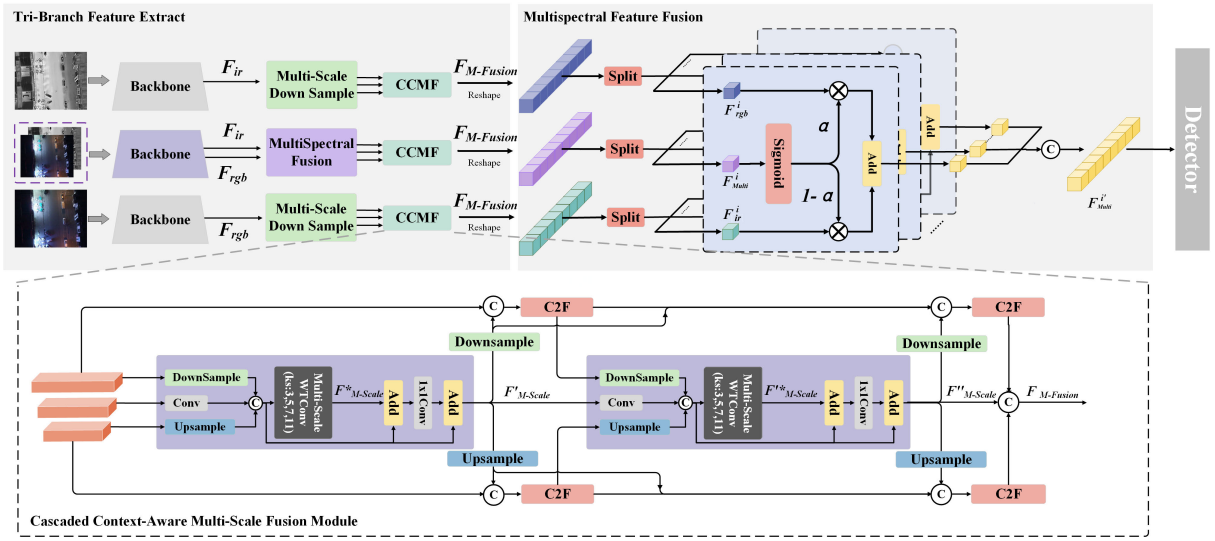


Fig. 2. Main architecture of our tribranch cascaded context-aware fusion network. It incorporates a CCMF module to mitigate the loss of small-object features, and a MFF.

of three branches: two branches process RGB and IR images separately; the third branch receives both RGB and IR images to globally perceive information and generate fusion weights for the features extracted by the other two branches. Each branch includes a feature extraction component and a CCMF module for gradual enhancement. Below, we present the details of each module.

A. Multibranch Feature Extract

Our method introduces a three-branch network specifically designed to enhance feature information through diversified multispectral representation learning.

First, we utilize two branches dedicated to single-spectral processing, which independently obtain image features F_{rgb} and F_{ir}, followed by downsampling to extract multiscale representations. Consistent with prior works [1], [27], we select features at the 1/8, 1/16, and 1/32 spatial scales as inputs for subsequent modules. These three scales correspond to the commonly used P3, P4, and P5 feature levels, respectively. Let s_i denote the downsampling rate corresponding to feature level i, where $s_{\text{P3}} = 8$, $s_{\text{P4}} = 16$, and $s_{\text{P5}} = 32$. We denote the downsampling operator for the RGB branch as $\mathcal{D}_{\text{rgb}}$ and for the IR branch as $\mathcal{D}_{\text{ir}}$. Applying these operators to the initial features yields multiscale features for each spectrum

$$F^i_{\text{rgb}} = \mathcal{D}_{\text{rgb}}\left(F_{\text{rgb}}, s_i\right), \quad i \in \{\text{P3}, \text{P4}, \text{P5}\} \tag{1}$$

$$F^i_{\text{ir}} = \mathcal{D}_{\text{ir}}\left(F_{\text{ir}}, s_i\right), \quad i \in \{\text{P3}, \text{P4}, \text{P5}\}. \tag{2}$$

Here, F^i_{rgb} and F^i_{ir} represent the multiscale features of the RGB and IR branches at level i, respectively.

The third branch simultaneously receives both spectral modalities as input. To demonstrate the effectiveness of our proposed method, we adopt the simplest fusion strategy, as illustrated in Fig. 3, performing elementwise addition on the multiscale feature pairs. The fusion process is formulated as follows:

$$F^i_{\text{fuse}} = \text{ConAdd}\left(F^i_{rgb}, F^i_{ir}\right), \quad i \in \{\text{P3}, \text{P4}, \text{P5}\}. \tag{3}$$

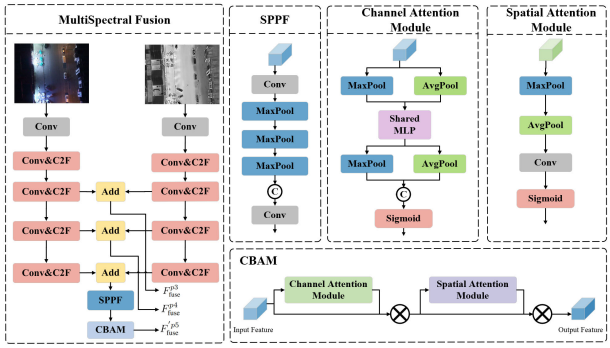


Fig. 3. Structural diagram of the multispectral feature fusion module. We perform a straightforward fusion of features across different scales, and additionally apply a CBAM module to the deep feature map (P5) to enhance the learning of finer-grained semantic features.

Finally, the F_{p5} feature undergoes further refinement through an additional feature extraction module, expressed as

$$F^{'p5}_{\text{fuse}} = \text{CBAM}\left(\text{SPFF}\left(F^{P5}_{\text{fuse}}\right)\right). \tag{4}$$

In particular, the architectural configuration of the convolutional block attention module (CBAM) [33] follows the original design specifications in the seminal work. Meanwhile, the spatial pyramid pooling-fast (SPPF) module retains the inherent structural integrity and operational principles of the YOLOv8 framework.

The above features are then fed into the CCMF module within each branch.

B. Cascaded Context-Aware Multiscale Fusion Within Each Branch

Usually, as a network deepens, the features of small objects gradually weaken, leading to missed detections. To address this, we not only adopt a multibranch network architecture but also introduce a CCMF module in each branch to mitigate

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the risk of losing small-object information. Fig. 2 illustrates the detailed design of our CCMF module.

We take the multiscale features extracted in Section III-A as input, denoted as $\left\{F_m^j \mid j \in \{1/8, 1/16, 1/32\}, m \in \{rgb, ir, multi\}\right\}$.

First, we perform a resolution alignment on these multiscale features

$$F_m'^{1/8} = \text{Adown}\left(F_m^{1/8}\right) \tag{5}$$

$$F_m'^{1/16} = \text{Conv}\left(F_m^{1/16}\right) \tag{6}$$

$$F_m'^{1/32} = \text{Conv}\left(\text{Upsample}\left(F_m^{1/32}\right)\right) \tag{7}$$

where Adown is a down-sampling operation [34]. The convolution operation employs a kernel size of $k = 1$, stride $s = 1$, and padding size $p = \lfloor k/2 \rfloor$, followed by batch normalization and SiLU activation. The aligned features are input into the multiscale WTConv [35] module.

In the multiscale WTConv module, a concatenation operation is first performed on its multiscale features to obtain a stacked multiscale representation $F_{M-Scale}$.

Next, $F_{\text{M-Scale}}$ is divided into multiple scales again using wavelet transform convolution (WTConv) with kernel sizes 5×5, 7×7, 9×9, and 11×11, respectively, leading to features $F_{\text{M-Scale}}^k$, where k represents WTConv at different scales

$$F_{\text{M-Scale}} = \text{Concat}\left(F_m^{j'}\right) \tag{8}$$

$$F_{\text{M-Scale}}^k = \text{WTConv}_k\left(F_{\text{M-Scale}}\right), \quad k \in \{5, 7, 9, 11\}. \tag{9}$$

Finally, the generated multiscale features are spatially aligned through a padding operation and subsequently aggregated. The aggregated features are further fused with the original input through residual connections, as illustrated at the bottom of Fig. 2, producing the enhanced feature representation $F'_{\text{M-Scale}}$

$$F_{\text{M-Scale}}^* = \sum_{k\in\{5,7,9,11\}} \text{Pad}_{p=\lfloor \frac{k}{2} \rfloor}\left(F_{\text{M-Scale}}^k\right) \tag{10}$$

$$F'_{\text{M-Scale}} = F_{\text{M-Scale}} + \text{Conv}\left(F_{\text{M-Scale}} + F_{\text{M-Scale}}^*\right). \tag{11}$$

The convolution operation employs a kernel size of $k = 3$, stride $s = 2$, and padding size $p = \lfloor k/2 \rfloor$, followed by batch normalization and SiLU activation, to preserve the spatial alignment of feature maps generated by WTConv layers with different kernel sizes.

Then, $F'_{\text{M-Scale}}$ is processed through upsampling and down-sampling operations to generate features at different scales. These features are concatenated with the original features at the corresponding scales and subsequently passed through the C2F [36] module for feature enhancement. We repeat this process once and finally obtain the output feature $F_{\text{M-Fusion}}$.

The input features are divided into N parts, with each feature represented as $(Z_m^i)_{i=1}^{\text{N}} \in \mathbb{R}^{H\times W\times \text{C/N}}$, where $m \in \{rgb, ir, multi\}$.

Here, H, W, and C denote the height, width, and channel dimension of the feature maps, respectively, while N represents the number of channel groups used for feature partitioning. In all experiments, we set $N = 8$ to balance representation capacity and computational efficiency. α is computed in a groupwise manner via a sigmoid function applied to each fused feature group

$$\alpha = \text{sigmoid}\left(Z_{Multi}^i\right) \tag{12}$$

where $\alpha \in \mathbb{R}^{H\times W\times C/N}$ denotes a spatial-channel attention map and is generated independently for each feature group

$$Z_{Multi}^i = \alpha Z_{rgb}^i + (1-\alpha) Z_{ir}^i \tag{13}$$

$$F_{Multi}^i = \text{Concat}\left(Z_{Multi}^1, Z_{Multi}^2, \ldots, Z_{Multi}^N\right). \tag{14}$$

In particular, the sigmoid operation generates adaptive fusion weights for each spatial location and channel dimension, enabling the network to dynamically balance RGB and IR feature contributions under different illumination conditions and scene complexities.

IV. Experiments

In this section, we first introduce the evaluation datasets, performance metrics, and implementation details adopted in the experiments. Subsequently, we present quantitative experimental results on each target dataset and conduct a differentiated comparative analysis against the current state-of-the-art methods specific to each dataset (as the state-of-the-art methods vary across different datasets). Finally, ablation studies are performed on the large-scale DroneVehicle and LLVip datasets to validate the effectiveness of the proposed method.

A. Evaluation Dataset

In the experimental dataset section, we adopt four widely used multispectral remote sensing datasets to verify the effectiveness and generalizability of the proposed method in multispectral tasks. These four datasets are: DroneVehicle (a dense small-object dataset), VEDAI (a small-object dataset), RGBT-Tiny (a tiny object dataset), and LLVip (a pedestrian dataset mainly consisting of medium and large objects). A brief introduction to these datasets is provided below.

The **DroneVehicle** [19] dataset is a publicly available dataset focused on multispectral object detection and recognition, particularly for vehicle detection tasks in nadir-view scenes. The dataset comprises 56 878 images, with half being RGB images and the other half IR images. It includes a total of 953087 annotations, consisting of 500517 infrared image annotations and 452570 RGB image annotations. It consists of high-resolution RGB and IR images captured by drones, featuring various types of vehicles, such as cars, trucks, and motorcycles, under different weather and lighting conditions. We will adopt the same evaluation method as used in previous articles, utilizing this dataset, dividing the dataset into standard training, validation, and test sets in a 6:1:3 ratio.

The **VEDAI** [37] (Vehicle Detection in Aerial Imagery) dataset is designed for object detection in aerial imagery, specifically focusing on vehicles. There are 1210 images and 3700 instances in the dataset. It consists of high-resolution multispectral images, including both RGB and IR channels, allowing for robust detection of objects under various environmental conditions. The dataset includes a wide variety of vehicle types, such as cars, trucks, and planes, captured from different angles and altitudes. VEDAI is commonly used to

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benchmark object detection models, especially in the context of multispectral fusion, and provides valuable challenges for small-object detection tasks. To ensure fairness, we adopted a ten-fold cross-validation approach for evaluation on this dataset, consistent with other methods that utilize the same dataset. The ratio of the training set to the test set is 9:1.

The **RGBT-Tiny** [30] dataset is a publicly available benchmark designed for visible-thermal tiny object detection. It contains 115 aligned RGB-thermal sequences, comprising about 93 000 frames and more than 1.2 million bounding-box annotations. The dataset covers seven object categories across eight different scenarios, featuring large variations in illumination, density, and environmental conditions. A notable characteristic is that over 81% of the annotated targets are smaller than 16×16 pixels, making the dataset highly challenging for detection tasks. Following prior works that utilize this dataset, we also adopt the standard data split provided by the authors for training, validation, and testing.

The **LLVip** [16] is a low-light vision RGB-IR image pair dataset containing 30 976 images, or 15 488 pairs, across 24 nighttime scenes and two daytime scenes. We selected this dataset to validate the generalization ability of our proposed method, demonstrating that it can not only be applied to multispectral dense target and small-object tasks but also effectively detect large objects. The dataset primarily features pedestrian targets with relatively large pixel proportions, allowing for a comprehensive assessment of our method's detection performance in various visibility scenarios, particularly in challenging nighttime environments. We will adopt the same evaluation method as used in previous articles, utilizing this dataset, dividing the dataset into standard training and test sets in an 8:2 ratio.

B. Evaluation Metrics

We adopt average precision (AP) and mean AP (mAP) to evaluate multispectral object detection performance. The intersection over union (IoU) between the predicted box B_{pred} and ground truth box B_{gt} is defined as

$$\text{IoU} = \frac{|B_{\text{pred}} \cap B_{\text{gt}}|}{|B_{\text{pred}} \cup B_{\text{gt}}|}. \tag{15}$$

Based on IoU, precision and recall are computed as

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

$$\text{Recall} = \frac{TP}{\text{Total Ground Truth}}. \tag{17}$$

AP is obtained from the precision-recall curve

$$\text{AP} = \int_0^1 \text{Precision}\,(\text{Recall})\; d\text{Recall}. \tag{18}$$

The metric at IoU threshold 0.5 is denoted as AP_{50}, while mAP is defined as

$$\text{mAP} = \frac{1}{N} \sum_{c=1}^{N} \text{AP}_c \tag{19}$$

where N denotes the number of categories.

For the RGBT-Tiny dataset, SAFit is additionally adopted following prior works:

$$\text{SAFit} = \frac{1}{1 + e^{-\left(\sqrt{A}/C - 1\right)}} \times \text{IoU} + \left(1 - \frac{1}{1 + e^{-\left(\sqrt{A}/C - 1\right)}}\right) \times \text{NWD(C)} \tag{20}$$

where A denotes the area of the ground-truth box, C is a scale balance constant, and NWD represents the normalized Wasserstein distance [47].

C. Implementation Details

We conducted our experiments using an Nvidia RTX 4090 GPU. We implemented our model architecture with PyTorch 2.3.1 and CUDA 12.2. For the DroneVehicle dataset, the input image size is 640×640, whereas for the VEDAI dataset, it is 1024×1024. The batch size for both datasets is set to 16, and the training is carried out for 200 epochs. We employed stochastic gradient descent (SGD) as the optimizer, with a momentum of 0.937 and a weight decay of 0.002. We will release our code. In the testing phase, the batch size is set to 1, and the confidence threshold for visualized results is set to 0.5.

D. Quantitative Results

1) Results on DroneVehicle: Table I presents the comparison of results between our method and other state-of-the-art methods on the DroneVehicle dataset. The experiment is divided into three sections: two single-spectral detection sections and one multispectral detection section. In this table, our single-spectral detection method is based on our base model with the introduction of the CCMF module. In the RGB spectrum, compared to the second-best, our method improves performance on the categories of car, truck, freight car, bus, van, and mAP@50 by 1.69%, 11.77%, 2.26%, 2.86%, 12.81%, and 8.69%, respectively. In the infrared spectrum, our method outperforms the second-best on the categories of car, truck, freight car, bus, van, and mAP@50 by 8.30%, 12.60%, 8.20%, 3.40%, 13.33%, and 8.00%, respectively. This demonstrates the effectiveness of our method in small-object detection. In the multispectral section, compared to the second-best method, our method improved performance on the categories of car, truck, Freight car, bus, van, and mAP@50 by 0.76%, 5.25%, 1.51%, 1.72%, 0.33%, and 3.47%, respectively. Furthermore, using mAP@50 as the evaluation metric, the precision of our method on multispectral data exceeded that of single-spectrum data by 4.67%–2.38%, respectively, proving the superiority of our method in addressing the semantic conflicts of multispectral information. Overall, the results on the DroneVehicle dataset clearly demonstrate the strong adaptability and robustness of our approach. TCFNet consistently surpasses competing methods in all major categories and achieves the highest mAP@50, further validating the effectiveness of our fusion strategy and CCMF module in handling small-object detection under complex remote sensing conditions.

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TABLE I

RESULTS ON THE DRONEVEHICLE TEST SET. THE EVALUATION METRIC FOR EACH CATEGORY IS AP@50. AMONG THEM, THE VERSION WITH OBB REPRESENTS THE ORIENTED BOUNDING BOXES (OBBS) METHOD. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Spectral	Venue	Year	Car	Truck	Freight car	Bus	Van	mAP@50
RetinaNet(OBB) [38]	RGB	ICCV	2017	67.50	28.24	13.72	62.05	19.26	38.10
YOLOv5(OBB) [39]		-	2020	89.29	59.89	49.51	86.72	45.06	66.10
TSFADet [11]		ECCV	2022	89.47	65.56	60.36	89.63	52.82	71.57
YOLOv8(OBB) [36]		-	2023	95.90	70.10	50.80	93.70	50.30	72.16
PKINet-S [40]		CVPR	2024	76.40	53.00	40.80	79.10	45.70	59.00
TCFNet_CCMF(Ours)		-	2025	**97.59**	**81.87**	**62.62**	**96.56**	**65.63**	**80.85**
RetinaNet(OBB) [38]	IR	ICCV	2017	79.86	32.84	28.05	67.32	16.44	44.90
YOLOv5(OBB) [39]		-	2020	89.29	59.89	49.50	86.72	42.06	65.50
TSFADet [11]		ECCV	2022	89.78	66.26	61.44	89.60	53.17	72.05
YOLOv8(OBB) [36]		-	2023	97.70	70.30	63.10	93.10	51.50	75.14
PKINet-S [40]		CVPR	2024	90.20	67.30	56.30	88.80	51.20	70.80
TCFNet_CCMF(Ours)		-	2025	**98.50**	**82.90**	**71.30**	**96.50**	**66.50**	**83.14**
Halfway Fusion [41]	Multi	arXiv	2016	89.85	60.34	55.51	88.97	46.28	68.19
CIAN(OBB) [42]		IF	2019	89.98	62.47	60.22	88.90	49.59	70.23
AR-CNN(OBB) [43]		CVPR	2022	90.08	64.82	62.12	89.38	51.51	71.58
UA-CMDet [19]		TCSVT	2022	87.51	60.70	46.80	87.08	37.95	64.01
TSFADet [11]		ECCV	2022	89.88	67.87	63.74	89.81	53.99	73.06
CALNet [1]		ACM MM	2023	90.30	76.15	62.97	89.11	58.46	75.40
SLBAF-Net [44]		CVPR	2024	90.20	72.00	68.60	89.90	59.90	76.10
ODAF [44]		CVPR	2024	90.30	76.80	73.30	90.30	66.00	79.40
DMM [45]		TGRS	2025	90.40	79.80	68.20	89.90	68.60	79.40
ADMPF [46]		TGRS	2025	97.94	82.28	69.34	95.93	64.77	82.05
MDPMFN [22]		TGRS	2025	97.95	80.87	67.68	95.50	64.33	81.26
TCFNet_Full(Ours)		-	2025	**98.71**	**87.53**	**74.81**	**97.65**	**68.93**	**85.52**

TABLE II

RESULTS ON THE VEDAI TEST SET. THE EVALUATION METRIC FOR EACH CATEGORY IS AP@50. AMONG THEM, THE VERSION WITH OBB REPRESENTS THE OBBS METHOD. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Spectral	Venue	Year	Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	mAP@50
Faster R-CNN [48]	RGB	NeurIPS	2015	71.40	67.60	70.50	54.20	58.60	61.00	52.30	59.50	61.50
RetinaNet [38]		ICCV	2017	48.90	37.50	21.40	16.80	17.65	15.90	4.40	5.90	20.70
RoITransformer [49]		CVPR	2019	77.30	71.50	73.60	56.10	64.25	64.70	56.70	60.20	65.40
S^2ANet [50]		TGRS	2021	74.50	65.10	61.70	47.30	23.45	55.60	16.70	32.50	44.50
Oriented R-CNN [51]		ICCV	2021	77.60	72.30	76.70	59.70	63.80	62.80	60.10	60.90	66.40
CMAFF [52]		PR	2022	91.10	82.30	75.10	78.30	33.30	81.20	**71.80**	62.20	74.30
YOLOv8 [36]		-	2023	86.89	78.23	74.33	64.58	54.96	72.18	45.54	66.23	67.87
TCFNet_CCMF(Ours)		-	2025	**92.43**	**86.04**	**80.33**	**78.41**	**65.42**	**88.07**	68.62	**91.87**	**81.40**
Faster R-CNN [48]	IR	NeurIPS	2015	71.60	66.50	68.10	49.10	50.55	49.20	35.60	57.00	55.40
RetinaNet [38]		ICCV	2017	44.20	32.10	17.10	15.30	19.55	9.40	4.00	7.20	18.70
RoITransformer [49]		CVPR	2019	76.10	70.70	71.20	51.70	55.80	51.90	46.90	64.30	60.50
S^2ANet [50]		TGRS	2021	73.00	65.60	59.20	39.20	17.55	41.90	13.90	32.30	40.00
Oriented R-CNN [51]		ICCV	2021	77.00	72.20	73.60	55.00	55.05	47.50	49.40	63.20	60.90
CMAFF [52]		PR	2022	86.70	75.90	66.60	77.10	43.30	62.30	**70.70**	62.20	74.00
YOLOv8 [36]		-	2023	88.25	76.21	74.55	68.39	40.15	52.61	41.84	68.29	63.79
TCFNet_CCMF(Ours)		-	2025	**90.80**	**77.04**	**76.43**	**89.70**	**68.43**	**88.07**	52.61	**92.99**	**79.51**
CMAFF [52]	Multi	PR	2022	91.70	85.90	78.90	78.10	54.70	71.90	71.10	75.20	78.60
YOLOv8 [36]		-	2023	93.31	88.95	80.32	72.93	63.45	82.41	64.66	80.73	78.35
GHOST [53]		TGRS	2023	88.82	86.06	74.33	**86.96**	61.28	86.00	71.76	87.26	80.31
DMM [45]		TGRS	2025	84.20	72.30	79.00	65.70	**74.90**	72.30	**93.60**	72.50	75.00
ADMPF [46]		TGRS	2025	94.45	90.25	**84.74**	74.41	57.16	74.74	85.46	91.86	81.63
TCFNet_Full(Ours)		-	2025	**95.79**	**91.64**	83.63	83.69	71.22	**91.17**	73.52	**96.98**	**85.96**

2) Results on VEDAI: Table II presents the comparison of results between our method and other state-of-the-art methods on the VEDAI dataset. The experiment is divided into three sections: two single-spectral detection sections (RGB and IR) and a multispectral detection section. In this table, our single-spectral detection method is based on our base model with the introduction of the CCMF module.

In the RGB spectrum, compared to the second-best method, our method improves performance on the categories of car, pickup, camping, truck, other, tractor, and van by 1.33%, 3.74%, 3.64%, 0.11%, 10.46%, 6.87%, and 21.57%, respectively. In the boat category, our method achieved the second-best performance. For the overall detection metric mAP@50, our approach obtained the best result, with an improvement of 7.10% compared to the second-best method.

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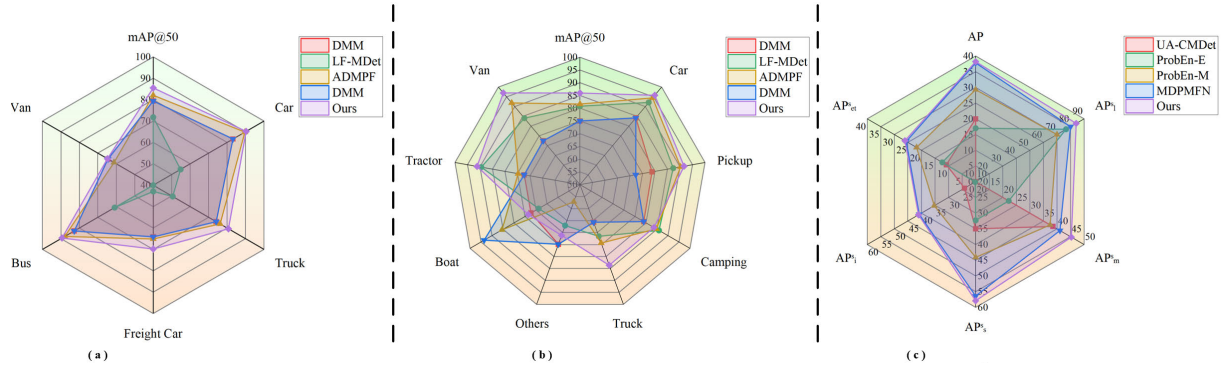


Fig. 4. Radar chart of detection results on three multispectral remote sensing datasets. (a) Drone vehicle dataset. (b) VEDAI dataset. (c) RGBT-tiny dataset.

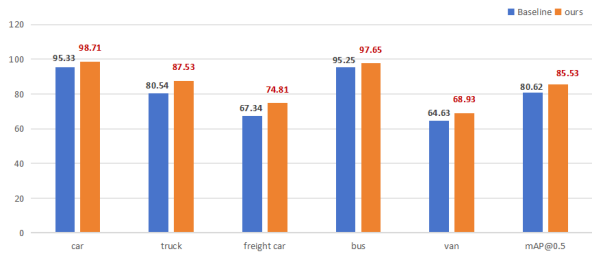


Fig. 5. Results compared with our base model [36] on the DroneVehicle test set.

TABLE III

COMPARISON OF FUSION METHODS ON THE RGBT-TINY DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	AP^s_{et}	AP^s_t	AP^s_s	AP^s_m	AP^s_l	AP
UA-CMDet [19]	10.8	24.1	35.0	38.5	9.7	20.0
ProbEn-E [30]	12.3	20.2	32.3	22.2	76.7	17.0
ProbEn-M	21.8	35.3	44.1	37.5	70.0	29.3
MDPMFN [22]	**25.6**	**40.8**	56.4	41.1	80.0	37.8
TCFNet(Ours)	24.8	40.7	**57.4**	**45.0**	**81.7**	**38.9**

TABLE IV

RESULTS ON THE LLVIP TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Venue Year	mAP@50	mAP@(50:95)
DM-Fusion [54]	TNNLS 2023	88.9	53.1
CALNet [1]	ACM MM 2023	-	63.4
MoE-Fusion [55]	ICCV 2023	91.0	-
MetaFusion [56]	CVPR 2023	91.0	56.9
TFNet [57]	TITS 2023	-	-
DDFM [58]	ICCV 2023	91.5	58.0
CSAA [59]	CVPR 2023	-	59.2
Diff-IF [60]	IF 2024	93.3	59.5
LENFusion [61]	TIM 2024	81.6	53.0
YOLO-Adaptor [62]	TIV 2024	96.5	-
CAMF [63]	TMM 2024	-	55.8
Fusion-Mamba [64]	arXiv 2024	96.8	62.8
MMPedestron [64]	ECCV 2024	-	**72.6**
MiPa [65]	arXiv 2024	98.2	66.5
TCF-Net (Ours)	-	**98.5**	66.1

In the infrared spectrum, compared with the second-best method, our approach achieved the best detection results in the categories car, truck, other, tractor, boat, and van, with improvements of 2.54%, 12.59%, 12.62%, 8.59%, and 21.57%, respectively. The remaining categories, pickup, camping, and boat, achieved the second-best results. For the overall detection metric mAP@50, our method achieved the best performance, with an improvement of 5.51% over the second-best method.

In the multispectral condition, compared with the second-best method, our approach achieved the best results in the categories car, pickup, tractor, and van, with improvements of 1.34%, 1.39%, 5.17%, and 5.12%, respectively. In the other categories, the truck and other categories achieved the second-best results, while the boat category achieved the third-best performance. For the overall detection metric mAP@50, our method obtained the best result, with an improvement of 2.53% compared to the second-best method. Overall, TCFNet exhibits strong and uniform advantages across all spectral conditions, achieving the highest mAP@50 scores. These results solidify the proposed method as a robust and generalizable solution for multispectral small-object detection in remote sensing.

3) Results on RGBT-Tiny: To verify the generalization ability of our method on datasets containing a large proportion of tiny objects, we conducted experiments on the RGBT-Tiny dataset. As shown in Table III and the radar chart in Fig. 4, our method achieves highly competitive performance. In particular, for extremely tiny and tiny objects, our results are very close to the current state-of-the-art, while for small, medium, and large objects, as well as the AP metric, our method surpasses existing approaches by 1%, 3.9%, 1.7%, and 1.1%, respectively. This indicates that despite the high proportion of tiny objects, blurred target textures, and large cross-modal discrepancies in the RGBT-Tiny dataset, our model can still maintain stable detection performance under more stringent thresholds.These results further demonstrate that our fusion strategy not only enhances target detectability but also effectively improves cross-modal feature alignment and high-precision localization. Therefore, our method remains competitive in the highly challenging task of tiny-object detection and exhibits strong reliability and generalization ability in precise bounding-box localization.

4) Results on LLVip: Table IV presents a comparison of our TCFNet with state-of-the-art multispectral detec-

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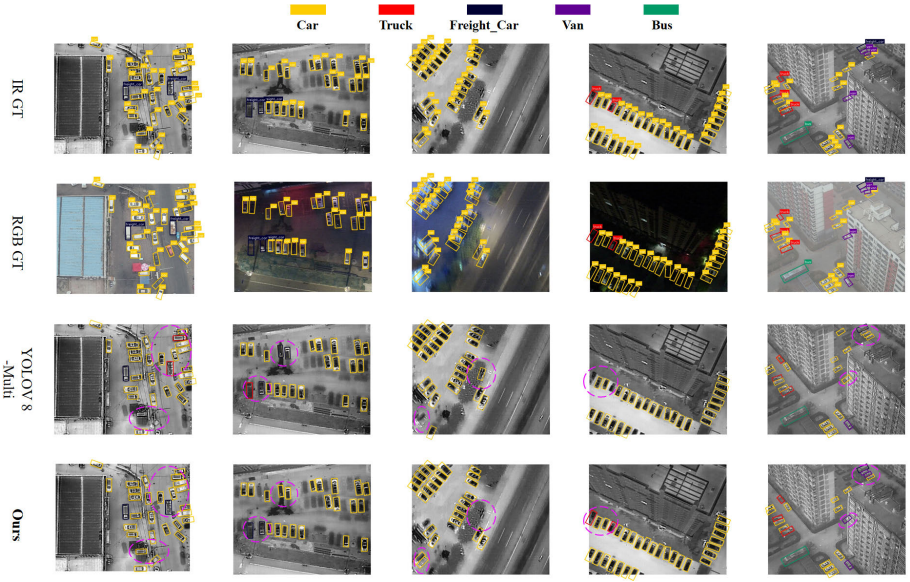


Fig. 6. Qualitative comparison of detection results on the DroneVehicle dataset. (First two rows) Display the ground truth for IR and RGB modalities. The regions highlighted by purple dashed circles demonstrate that our method effectively reduces missed detections and false alarms compared to the YOLOv8-Multi baseline, especially in dense parking lots and low-light environments.

TABLE V
ABLATION STUDY RESULTS ON THE DRONEVEHICLE AND LLVIP TEST SET

Dataset	Spectral	Methods	mAP@50	mAp@(50:95)	Parameters	GFLOPs
DroneVehicle	RGB	base	78.4	62.9	69.5M	263.9
		base+CCMF	80.8	64.4	75.9M	284.8
	IR	base	80.7	67.2	69.5M	263.9
		base+CCMF	83.2	68.7	75.9M	284.8
	RGB+IR	base	82.0	67.4	100.6M	396.9
		base+CCMF	84.3	68.7	107.1M	417.2
		base+CCMF+MFF	85.5	69.2	120.8M	426.7
LLVip	RGB	base	94.3	61.2	69.5M	263.9
		base+CCMF	95.2	61.9	75.9M	284.8
	IR	base	96.3	64.4	69.5M	263.9
		base+CCMF	97.1	65.2	75.9M	284.8
	RGB+IR	base	96.6	64.5	100.6M	396.9
		base+CCMF	97.4	65.1	107.1M	417.2
		base+CCMF+MFF	98.5	66.1	120.8M	426.7

tion approaches. Our method achieves an mAP@(50:95) of 66.1%, surpassing the method, CALNet, by 2.7%. In addition, it achieves the highest mAP@50 of 98.5%, outperforming the best method, MiPa, by 0.3%. These results demonstrate the effectiveness of our approach in improving multispectral detection accuracy, particularly in enhancing precision while maintaining competitive overall performance.

The proposed method, TCFNet, achieves outstanding performance on two major remote sensing datasets, DroneVehicle and Vedai, demonstrating its effectiveness in the remote sensing domain. To further evaluate the generalization capability of our method, we conducted additional experiments on the RGBT-Tiny dataset, which contains extremely small targets, and the LLVip dataset, which includes large single targets. The results show that TCFNet maintains highly competitive performance on tiny targets while also exhibiting advantages on large targets, highlighting the strong generalization ability of the proposed approach.

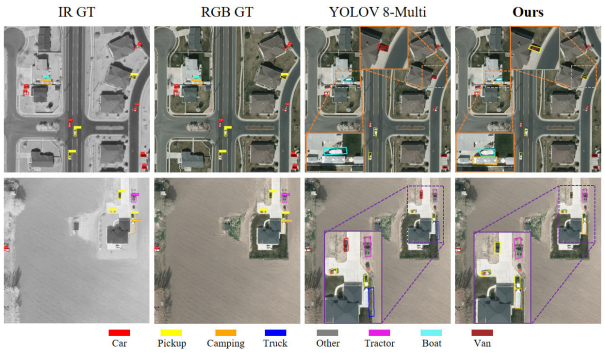


Fig. 7. Qualitative comparison of detection results on the VEDAI dataset.

E. Ablation Study

This section presents an ablation study of each proposed module on two datasets, with results shown in Table V.

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TABLE VI

ABLATION STUDY OF THE TRIBRANCH ARCHITECTURE ON DRONEVEHICLE. R, I, AND M DENOTE RGB, IR, AND MULTIBRANCHES

Methods	R	I	M	mAP@50	mAP@(50:95)
Method 1	✓		✓	84.8	68.9
Method 2		✓	✓	85.1	69.0
Method 3			✓	84.3	68.7
Ours	✓	✓	✓	**85.5**	**69.2**

TABLE VII

PERFORMANCE AND COMPLEXITY COMPARISON ACROSS DIFFERENT MODEL SCALES

Type	mAP50	mAP(50:95)	Param (M)	GFLOPs
n	74.9	59.4	**5.9**	**14.9**
s	78.1	63.2	19.3	53.3
m	81.6	65.7	45.0	135.0
l	83.4	67.8	74.4	276.0
x	**85.5**	**69.2**	120.8	426.7

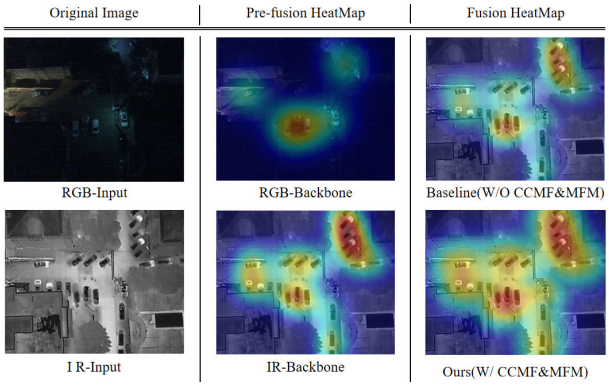


Fig. 8. Visualization results of the heatmap on DroneVehicle datasets.

From the results, it can be seen that the introduction of the CCMF module based on the YOLOv8 [36] base model has improved performance under various spectral conditions. As shown in Fig. 5, on the DroneVehicle dataset, which primarily consists of small targets, our CCMF module achieved an average performance improvement of 2.4% across different spectral conditions. On the LLVip dataset,

which mainly contains large targets, it achieves an average improvement of 0.8%. This demonstrates that the CCMF module performs particularly well for small-target tasks and also maintains favorable performance for large objects. Meanwhile, the model complexity is only modestly increased by 6.4M parameters and 20.9 Gflops in computation. These results indicate that the CCMF module enhances multispectral target detection accuracy without significantly increasing model complexity.

Compared to only applying CCMF, the MFF module brings a further performance increase of 1.1%–1.2%, with an additional 13.7M parameters and 20.9 Gflops in computation. Without the MFF method, the multispectral model showed no significant advantage over the infrared-only model. Therefore, the experimental results confirm that the MFF method effectively mitigates target feature conflicts between different spectra. Furthermore, our method achieves considerable performance improvements with only a moderate increase in computational complexity.

Overall, the proposed model demonstrates an impressive performance boost over the baseline model under multispectral conditions, achieving performance gains of 3.5% and 1.9% on the two datasets, respectively, with an additional 20.2M parameters and 29.8 Gflops.

Table VI summarizes the ablation results of the proposed multibranch architecture on the DroneVehicle dataset. Compared to the single-branch fusion methods (Method 3, 84.3% mAP@50), incorporating independent RGB (Method 1) and IR (Method 2) branches yields improved performances of 84.8% and 85.1%, respectively. Ultimately, our full architecture (Ours) achieves peak performance with 85.5% mAP@50 and 69.2% mAP@(50:95). These results demonstrate that the fine-grained textures from the RGB branch, sharp object contours from the IR branch, and dynamic weight allocation from the multibranch structure are all indispensable for robust detection, effectively addressing the complexities of multispectral data in drone-view scenarios.

Table VII compares the performance and complexity of TCFNet across various scales, demonstrating its excellent scalability. The lightweight version 'n' achieves 74.9% mAP50 with only 5.9M parameters, showing significant potential for edge deployment, while version 'x' pushes the detection boundary to 85.5% mAP50. In conclusion, TCFNet offers a flexible tradeoff between efficiency and accuracy, maintaining robust feature modeling capabilities across a wide range of parameter scales.

F. Qualitative Results

This section evaluates the effectiveness of TCFNet through qualitative comparisons. First, Fig. 6 illustrates detection results on the DroneVehicle dataset under extreme degradations, such as nighttime, blur, low illumination, and haze. While the baseline struggles with severe missed detections and localization errors in noisy regions (marked by purple dashed circles), TCFNet maintains robust feature representations, achieving high consistency with the ground truth even in near-dark conditions.

Furthermore, evaluations on the VEDAI dataset shown in Fig. 7, which is characterized by small objects and complex backgrounds, reveal similar trends. The baseline model exhibits significant limitations in suburban and urban areas marked by orange and purple boxes, leading to frequent misses of small vehicles and boats. On the contrary, TCFNet effectively mitigates these issues, demonstrating superior robustness and localization accuracy in challenging remote sensing scenarios.

Furthermore, the effectiveness of the proposed modules is quantitatively validated through heatmaps and feature maps. As shown in Fig. 8, the RGB backbone exhibits severe attention drift, overlooking dark and small targets, while the infrared backbone's attention remains scattered. Comparison of postfusion heatmaps reveals that the baseline (w/o CCMF

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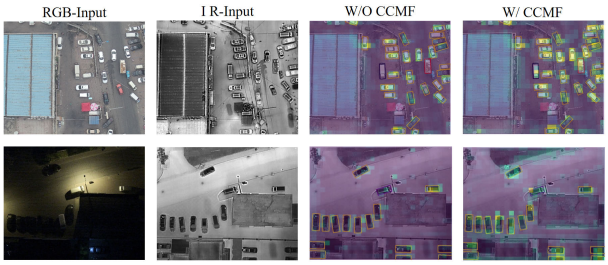


Fig. 9. Visualization results of the feature map on DroneVehicle datasets. Different stages represent different feature map phases.

and MFF) fails to integrate cross-modal complementary information, resulting in incomplete target activation. On the contrary, TCFNet, equipped with these modules, achieves optimized attention distribution, precisely focusing on all vehicles with consistent activation intensity. Feature maps in Fig. 9 further demonstrate the superiority of the CCMF module in suppressing background noise and enhancing feature extraction for small objects.

In summary, by progressively aggregating semantic and spatial information through the CCMF module and refining branch predictions via the MFF module, TCFNet demonstrates strong robustness against complex backgrounds and small-object challenges, confirming its superior capability in processing multisource heterogeneous data.

V. Conclusion

We propose a tribranch cascaded context-aware fusion network for multispectral small-object detection, designed to address the issue that the nature of the noise and the information varies significantly across frequency bands. Experimental results demonstrate that our multibranch design, the proposed within-branch cascaded context-aware multiscale fusion, and our branch-level feature fusion together effectively mitigate the issues of noise and information differences across frequency bands as well as small-object feature loss, achieving excellent performance on four datasets. Our approach achieves the best mAP@50 scores with up to a 3.5% improvement on four widely used benchmarks, the DroneVehicle, VEDAI, RGBT-Tiny, and LLVip datasets, compared to several state-of-the-art methods. This further confirms the effectiveness and generalization capability of our proposed method.

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